

Geometry In Practice and Numbers In Theory

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1 Introduction

The authors have for some years been studying the number theory which arises from a certain paper-folding algorithm. The work originated when one of the authors (JP) became involved in a local campaign in Northern California to reinvigorate the teaching of geometry. The plan which she conceived was to demonstrate a systematic procedure for folding paper strips to construct arbitrarily good approximations to certain regular convex polygons, and also to use the folded straight strips to construct interesting polyhedra – for example, the Platonic solids (see [9] for details about these constructions).

The campaign flourished – and the mathematics became increasingly intriguing. The first and obvious question was: which polygons can be constructed by the paper-folding algorithm? Now this algorithm may be described by a sequence of “down-folds” at the top of the strip, and “up-folds” at the bottom of the strip. It turns out that if we confine ourselves (as we, in fact, originally did) to a repeating pattern of down-folds and up-folds – say, m down-folds followed by n up-folds – and if we wish the top of the folded tape to describe the polygon, then our algorithm enables us to construct any s -gon, where s is an integer of the form

$$\frac{2^{m+n} - 1}{2^n - 1} \quad (1.1)$$

Thus our first number-theoretical questions are: When is (1.1) an integer? How may we recognize whether an integer s has the form (1.1), and if so, how may we determine m and n as functions of s ?

It turns out that (1.1) is an integer if and only if $n|m$, so that it is natural, and useful, to change notation and consider integers s given by

$$s = \frac{2^{xy} - 1}{2^x - 1}, \text{ for } x, y \text{ integers, } x \geq 1, y \geq 2 \quad (1.2)$$

The number theory, however, does not recognize the special role which the geometry assigns to the number 2. Thus, if we liberate ourselves by ignoring the geometer’s stubborn insistence that it is easiest to *bisect* angles by folding, we see that it is more informative (and requires no more effort) to consider integers of the form

$$s = \frac{t^{xy} - 1}{t^x - 1}, \text{ for } x, y \text{ integers, } x \geq 1, y \geq 2 \quad (1.3)$$

where $t \geq 2$ is a fixed positive integer. We call such numbers s (as defined by 1.3) *t-folding numbers*, and write F_t for the set of such numbers. They turn out to have fascinating properties; some elementary, some less so.

Now, if (1.1) is not an integer, then it is equivalent to a reduced fraction $\frac{b}{a}$, and our algorithm then enables us to fold regular star $\{\frac{b}{a}\}$ -gons in the terminology of Coxeter [1].

A regular star $\{\frac{b}{a}\}$ -gon is a closed sequence of b edges that visit, in order, every a th vertex ($\text{mod } b$) of a bounding regular convex b -gon. We include among the regular star polygons, the spiral case of the regular convex b -gon obtained when $a = 1$.

Once again, we are led to many questions: Which reduced fractions $\frac{b}{a}$ are representable as (1.1), and how do we determine m and n as functions of b and a ?

We are also led to study, for a fixed t , rational numbers of the form

$$\frac{t^p - 1}{t^q - 1}, \text{ for } p, q \text{ positive integers} \quad (1.4)$$

We announced many results relating to t -folding numbers and their rational analogs (1.4) in [3,4]. In Section 3, we collect together the most interesting results, and supplement them.

Returning to our paper-folding theme, it transpired that an obvious generalization of our “down m , up n ” procedure (which may be said to have period 2) enabled us to fold *all* regular star $\{\frac{b}{a}\}$ -gons, provided a, b are odd. The generalization simply consists of replacing a period 2 folding instruction (m, n) by a folding instruction $(k_1, k_2, \dots, k_r)$ of arbitrary period r . As we show in Section 2, if r is odd, then we must duplicate the folding instruction in order to find a repeating pattern of equally spaced intervals along the top of the tape.

The number theory now again takes over, but the direction is quite different. If the folding instruction $(k_1, k_2, \dots, k_r)$ produces the convex b -gon – or, indeed, any star $\{\frac{b}{a}\}$ -gon – then $k = \sum_{i=1}^r k_i$ is the *quasi-order* of 2 mod b , and in fact,

$2^k \equiv (-1)^r \pmod{b}$. This theory, and its consequences, treated *seriatim* in the papers [5,6,7,8], is described in some detail in Section 4; thus, in this section, we also achieve the (less obvious) generalization in which 2 is replaced by an arbitrary integer $t \geq 2$. We also answer the question – analogous to one posed for period 2 folding procedures – of what star polygons are produced by a given folding procedure $(k_1, k_2, \dots, k_r)$. Further developments of the number theory, unrelated to the geometry, have been made in one direction in [8], and in another by Froemke and Grossman in [2].

In Section 2, we describe the paper-folding algorithm. This has two aspects. First, there is a simple universal algorithm for actually constructing from a straight strip of paper (with parallel edges) a given regular star $\{\frac{b}{a}\}$ -gon where b and a are mutually prime odd integers and $a < (\frac{b}{2})$. The procedure requires that the tape have, at equally spaced intervals along its top edge, crease lines making angles of $(\frac{a}{b})\pi$ and $2(\frac{a}{b})\pi$ with the top edge of the tape. We call the final assembly routine the *FAT-algorithm*, since it is a simple Fold And Twist procedure. The second aspect is the procedure for actually putting the desired crease lines into the paper; this is the procedure which leads to the number theory.

In Section 5, we describe how, perhaps by the introduction of secondary fold lines, we may achieve the final generalization to any regular star $\{\frac{b}{a}\}$ -gon; that is, we allow a or b to be even. This material has not been published elsewhere, although we have discussed the simple matter of constructing arbitrary convex polygons.

In the Appendix, we describe our recent generalization [10] of the results of Section 3.

We have included in this monograph most of the proofs of our results. However, we would like to suggest to the reader that before studying a proof, he or she make an attempt to obtain a proof for himself/herself. This advice does not depend on any special features of our presentation – it is simply (as the great topologist J. H. C. Whitehead explicitly pointed out) the best way to learn mathematics.

A second excellent rule in reading a piece of mathematics is to ask oneself what questions naturally arise. We have presented – as you will already have noticed – our material as our own response to natural questions. In postscript to this monograph, we propose some questions for the reader's consideration, but of course, this should not inhibit the reader from thinking of his or her own. The questions we propose are related to the relevant sections of our paper by an obvious numbering device.

The authors would like to thank Professor J. R. Boyd for giving them this opportunity to present their theory and practice together, in its natural sequence, so that the reader need not depend on a set of papers scattered through the literature. They would also like to thank him for his very courteous and efficient handling of their manuscript.

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2 Folding Regular Star Polygons

In this section, we describe a precise folding procedure, involving a straight strip of paper (with parallel edges), that may be used to make the top edge of the strip approximate (to any desired degree of accuracy) a regular star $\{\frac{b}{a}\}$ -gon where a and b are mutually prime odd integers and $a < (\frac{b}{2})$. The procedure is systematic, easy to execute, and leads to very surprising relationships in number theory.

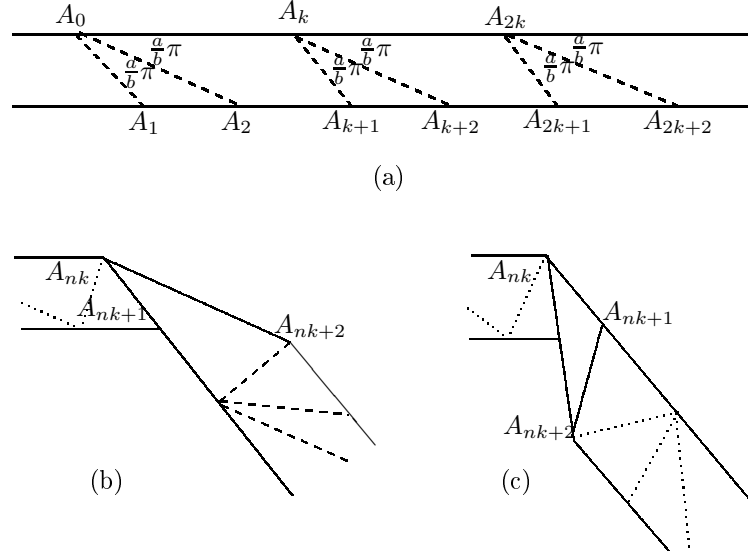


Figure 2.1

For the moment, assume that we have a straight strip of paper that has creases or folds along straight lines emanating from marked vertices at the top and bottom edges, and that those at the vertices A_{nk} , $n = 0, 1, 2, \dots, b$, on the top edge, form identical angles $(\frac{a}{b})\pi$, as shown in Figure 2.1(a). Suppose further that these vertices are equally spaced (we describe below how you might obtain such a strip). If we fold this strip on $A_{nk}A_{nk+2}$, as shown in Figure 2.1(b), and then twist the tape so that it folds on $A_{nk}A_{nk+1}$, as shown in Figure 2.1(c), the direction of the top edge of the tape will be rotated through an angle of $2((\frac{a}{b})\pi)$. We call this the process of *folding and twisting* the FAT-algorithm.

Now observe that if the FAT-algorithm is performed on a sequence of angles, each of which measure $(\frac{a}{b})\pi$, at the first b of a number of equally spaced locations along the top of the tape – in our case, at A_{nk} for $n = 0, 1, 2, \dots, b-1$ – then the top edge of the tape will have turned through an angle of $2a\pi$, so that the point A_{bk} will then be coincident with A_0 . Thus the top edge of the tape will have visited every a th vertex of a bounding regular convex b -gon, and hence determines a regular star $\{\frac{b}{a}\}$ -gon.

The conditions $a < (\frac{b}{2})$ fits beautifully with the geometry, since a regular star $\{\frac{b}{a}\}$ -gon is completely equivalent to a regular star $\{\frac{b}{(b-a)}\}$ -gon, where the tape visits the vertices in the opposite order.

We now describe a general procedure that can be used to obtain the desired crease lines on the strip of tape. We follow the description with an example.

Assume only that a, b are odd with $a < (\frac{b}{2})$, and that we wish to have a strip of paper on which the angle $(\frac{a}{b})\pi$ appears at regular intervals along the top edge. Designate the direction from left to right as the forward direction on the tape. Now begin folding by making an *initial* crease line in the downward forward direction (any downward sloping line will do!), from a point on the top edge of the tape that we label A_1 (for the purpose of describing it here), to a point A_2 at the bottom of the tape. (In fact, the first crease line does not even need to form an acute angle with the top edge of the tape in order to have the procedure work; it simply saves a little time if we take advantage of the fact that we know $(\frac{a}{b})\pi$ is acute.)

Now simply assume that the angle this initial line makes with the top edge is $(\frac{a}{b})\pi$. (In fact, if you execute the following instructions, a very good approximation of the desired angle will soon appear!) Call this desired angle the *putative* angle. Then continue to form new crease lines according to the following four Folding Rules:

- (1) The first new crease line emanates from the vertex A_2 .
- (2) Each new crease line goes in the forward direction along the strip of paper.
- (3) Each new crease line always *bisects* the angle between the last crease line and the edge of the tape from which it emanates.
- (4) The bisection of angles at any vertex continues until a crease line produces a putative angle of the form $(\frac{a'}{b})\pi$, where a' is an *odd* number; then the folding stops at that vertex, and commences at the intersection point of the last crease line with the other side of the tape.

Let us consider an example; say, $b = 7$, $a = 1$. It is clear that if we begin with an angle of $(\frac{1}{7})\pi$ at A_1 and adhere to the above rules, we will obtain a strip of tape with the angles and creases (indicated by dotted lines) shown in Figure 2.2. We call this the D^2U^1 folding procedure. This folding procedure is an example of a *primary* folding procedure of *period* 2.

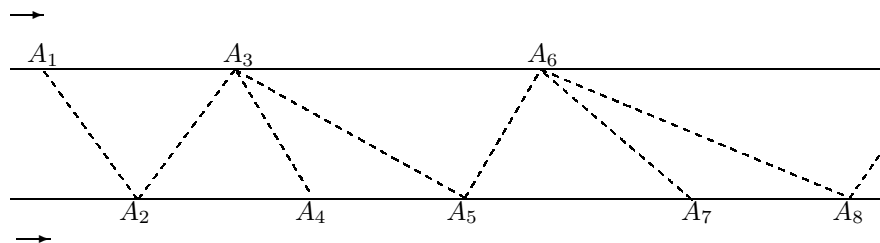


Figure 2.2

It is a surprising fact (for which we will indicate a line of proof later in this section) that starting with *any* putative angle $(\frac{a}{b})\pi$, where a, b satisfy our conditions, we will always obtain by our rules a primary folding procedure which produces angles that get closer and closer to the putative angle.

Furthermore, notice that in our particular case, starting with a putative angle of $\frac{\pi}{7}$ at the top of the tape produced a putative angle of $\frac{3\pi}{7}$ at the bottom of the tape. And, in fact, there are also pairs of angles, at equally spaced intervals, of $\frac{2\pi}{7}$ along the top edge of the tape. Thus, we may construct *all* of the regular star 7-gons from this tape. Here are the details.

Figure 2.3(a) shows how the D^2U^1 tape should look after a few folds (the first few triangles of the tape, which may have been irregular, have been discarded). If the FAT-algorithm is applied to the pairs of angles measuring $\frac{\pi}{7}$, at $A_9, A_{12}, \dots, A_{27}$, then A_{30} will coincide with A_9 , and the top edge of the tape will form a regular convex 7-gon (see Figure 2.3(d)).

Similarly, if the FAT-algorithm is applied to the pairs of angles measuring $\frac{2\pi}{7}$ at suitably spaced locations where they occur along the top edge of the tape in Figure 2.3(a), say at $A_9, A_{15}, \dots, A_{45}$, then the top edge of the tape can be “woven” through itself to form the regular convex $\{\frac{7}{2}\}$ -gon shown in Figure 2.4.

Finally, if the tape is flipped so that the bottom edge becomes the top, and the FAT-algorithm is applied to the pairs of angles measuring $\frac{3\pi}{7}$ at equally spaced locations that occur along the top, say at $A_{11}, A_{17}, \dots, A_{47}$, then the top edge of the tape will form a regular convex $\{\frac{7}{3}\}$ -gon, as shown in Figure 2.5(a), and when the excess at each point is folded under, the more conventional-looking star $\{\frac{7}{3}\}$ -gon of Figure 2.5(b) emerges.

Of course, if the vertices at which the FAT-algorithm is performed are selected far enough apart, it is possible to braid the tape through itself to form a $\{\frac{7}{3}\}$ -gon with a hole in the middle. However, just as in our illustration, even though the top edge would be delineating the polygon, there would be extra paper sticking out at each point (which could, of course, be folded around the point as was done in Figure 2.5(b)). (See Figures 2.4 and 2.5.)

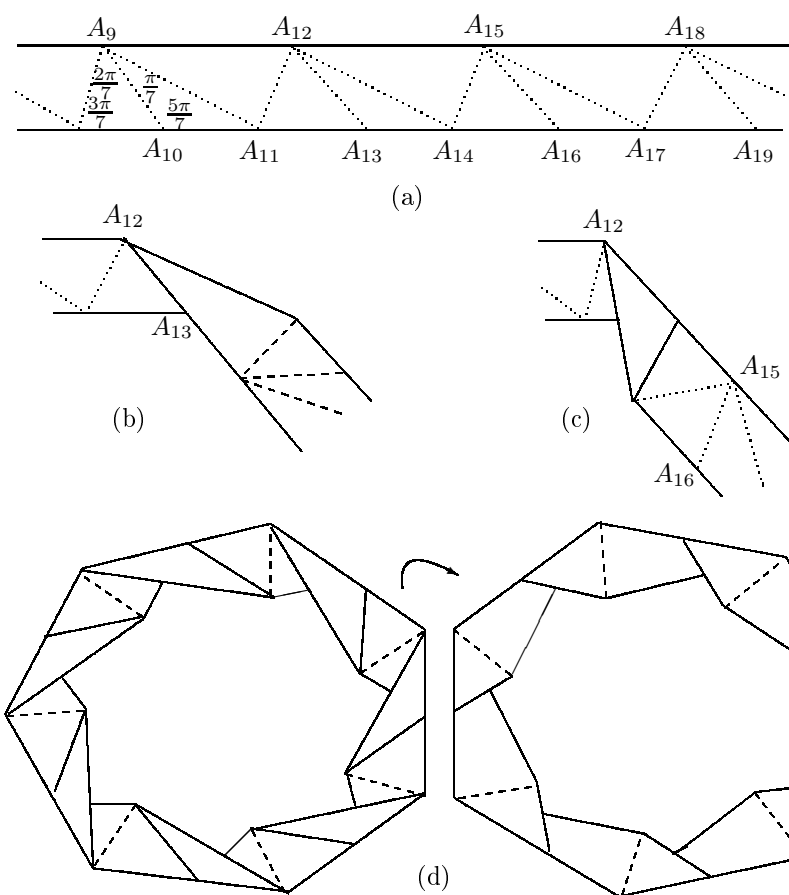


Figure 2.3

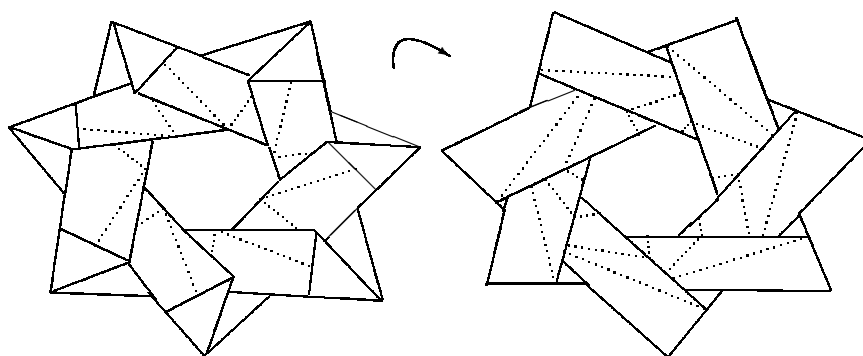


Figure 2.4

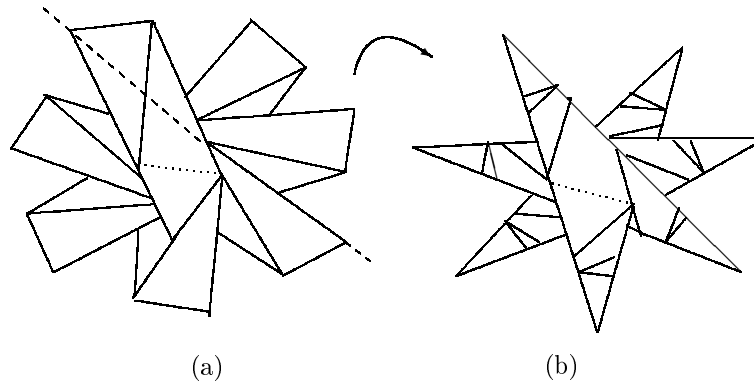


Figure 2.5

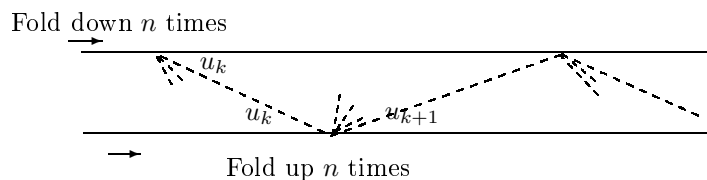


Figure 2.6

We are now in the lovely position of having many answerable questions. Here are just a few of the possibilities:

- (1) Can we prove in the above example, or in general, that the actual angles on the tape really converge to the putative angle we originally sought?
- (2) What happens if you consider general folding procedures with other periods, such as D^3U^3 , D^4U^2 , or $D^3U^1D^1U^3D^1U^1$? (The period is determined when the exponents repeat, so these examples have periods 1, 2, and 3, respectively.)
- (3) How many different regular convex polygons is it possible to fold with a 2-period D^mU^n folding procedure? Which ones are they? How do we recognize them? How do we determine the appropriate m and n in order to get them?
- (4) Is there an easiest folding procedure that will produce a regular convex b -gon for every odd number $b > 3$? If so, how do we find it?
- (5) How many of the regular star polygons is it possible to construct by folding paper?

The answers to all of these questions (and, in fact, many more) are *known*! In the remainder of this section, we will first indicate how to answer questions

1 and 2, and then illustrate a method for dealing with question 4. Question 3 is discussed in [4], and we show in the last section of this paper that the answer to question 5 is the best possible; that is, in theory, we can construct by folding paper, *all* regular star polygons. Furthermore (in answer to question 4), we indicate how the procedures we use give the best possible method of constructing the regular star polygons.

Let us begin to answer these questions. It turns out that the answers to questions 1 and 2 can be obtained by a straightforward use of the following:

Lemma 2.1: Let the sequence $\{x_k\}$, $k = 0, 1, 2, \dots$, be generated by the recurrence relation

$$x_k + ax_{k+1} = b, \quad k = 0, 1, 2, \dots \quad (2.1)$$

So if $|a| > 1$, then $x_k \rightarrow \frac{b}{(1+a)}$.

Proof: Set $x_k = \frac{b}{(1+a)} + y_k$. Then $y_k + ay_{k+1} = 0$. It follows that $y_k = \left(\frac{-1}{a}\right)^k y_0$.

If $|a| > 1$, $\left(\frac{-1}{a}\right)^k \rightarrow 0$, so that $y_k \rightarrow 0$ as $k \rightarrow \infty$. Hence $x_k \rightarrow \frac{b}{(1+a)}$.

This result may also be viewed as a special case of the well-known contraction mapping principle (see [11]). We point out that it is significant that neither the convergence nor the limit depends on the initial value x_0 . And, as we will soon demonstrate, the proof of the lemma tells us that the convergence of our folding procedure is rapid, since in all cases $|a|$ will be a power of 2.

Let us begin our discussion of convergence by looking at the general 1-period folding procedure $D^n U^n$. A typical portion of the tape would appear as illustrated in Figure 2.6. Then, if the folding process had been started with an arbitrary angle u_0 , we would have

$$u_k + 2^n u_{k+1} = \pi, \quad k = 0, 1, 2, \dots \quad (2.2)$$

with $|u_0| < \frac{\pi}{2}$. Equation (2.2) is of the form (2.1), so it follows immediately from Lemma 2.1 that

$$u_k \rightarrow \frac{\pi}{2^n + 1} \text{ as } k \rightarrow \infty \quad (2.3)$$

Furthermore, we can readily see that if the original fold differed from the putative angle of $\frac{\pi}{(2^n+1)}$ by an error E_0 , then the error at the k th stage of the folding process would be given by

$$E_k = \frac{E_0}{2^{nk}}$$

We see from (2.3) that the $D^n U^n$ folding procedure produces tape from which we may construct n regular convex $(2^n + 1)$ -gons. In fact, because the measure of the smallest acute angle at the bottom and the top of this tape is the same, we may construct the $(2^n + 1)$ -gons by special methods which are not as comprehensive as our FAT-algorithm. Figure 2.7 shows how $D^2 U^2$ tape may be folded along just the short lines of the folded tape to form the outline of a

small regular convex pentagon (a); and along just the long lines of the folded tape to form the outline of a slightly larger regular convex pentagon (b); and finally, in (c) we show the regular convex pentagon formed by executing the FAT-algorithm.

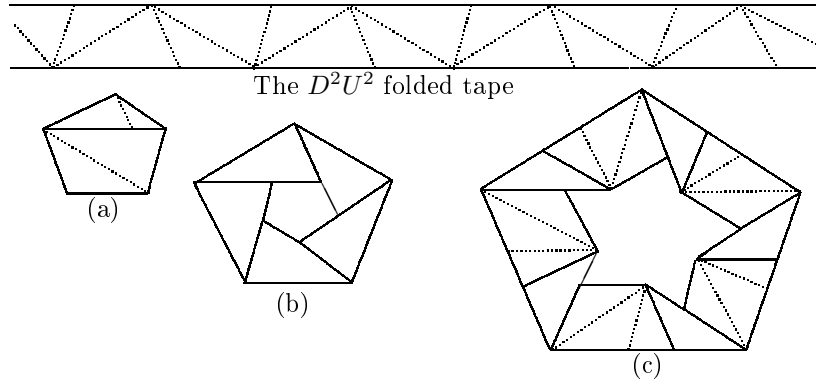


Figure 2.7

We feel certain that both the ancient Greeks and Gauss would have appreciated the fact that when $n = 1, 2, 3, 4, 8, 16, 32$, we obtain, in theory, easy constructions for regular 3, 5, 9, 17, 257, 65537, and $(2^{32} + 1)$ -gons, respectively.

The Greeks first raised (and solved) the question of how to construct (with straight-edge and compass) regular N -gons of the form $2^n N_0$, where $N_0 = 3, 5$, or 15 , and $n \geq 0$; while Gauss (1777-1855) proved that regular N -gons are constructible with a straight-edge and compass if and only if N is of the form $2^n N_0$ with N_0 a product of distinct Fermat primes; that is, a product of primes of the form $2^{2^m} + 1$.

We would argue that in practice, the approximations obtained by folding paper are more accurate than the real-world constructions with a straight-edge and compass – for the latter are only perfect in the mind. In both cases, what you actually get is a function of human skill, but the skill element in paper-folding is much less severe.

Now, suppose we look at the general 2-period folding procedure, $D^m U^n$. In this case, a typical portion of the tape would appear as shown in Figure 2.8.

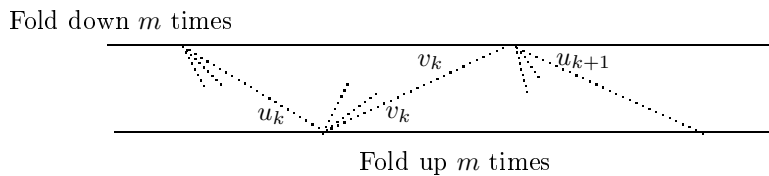


Figure 2.8

If the folding process had been started with an arbitrary acute angle u_0 at the top of the tape, we would have

$$\begin{aligned} u_k + 2^n v_k &= \pi \\ v_k + 2^m u_{k+1} &= \pi \end{aligned}$$

and hence, it follows that

$$u_k - 2^{m+n} u_{k+1} = \pi(1 - 2^n) \text{ with } u_0 < \frac{\pi}{2}, \quad k = 0, 1, 2, \dots \quad (2.4)$$

and thus, again using Lemma 2.1, we see that

$$u_k \rightarrow \frac{2^n - 1}{2^{n+m} - 1} \pi \text{ as } k \rightarrow \infty \quad (2.5)$$

so that the FAT-algorithm will produce from this tape a star $\{\frac{b}{a}\}$ -gon, with $b = 2^{m+n} - 1$, $a = 2^n - 1$. By symmetry, we can infer that

$$v_k \rightarrow \frac{2^m - 1}{2^{n+m} - 1} \pi \text{ as } k \rightarrow \infty \quad (2.6)$$

Furthermore, if we assume an initial error of E_0 , then we know that the error at the k th stage (when the folding $D^m U^n$ has been done exactly k times) will be given by $E_k = \frac{E_0}{2^{(n+m)k}}$. Hence, we see that in the case of our $D^2 U^1$ folding, any initial error E_0 is reduced by a factor of $\frac{1}{2^3}$ between consecutive appearances of any one of the putative angles.

It should now be clear why we advised throwing away the first part of the tape – but likewise, it should also be clear that it is never necessary to throw away very much of the tape. In practice, convergence is very rapid indeed, and if one made it a rule of thumb to always throw away the first 20 fold lines on the tape for any iterative folding procedure, it would turn out to be a very conservative rule ($\frac{(\pi/2)}{2^{20}}$ is a very small error!).

The above technique can be used to prove the convergence for any given folding sequence of general period, say r . This is fine, once we have identified the folding sequence, but the question that is more likely to arise for the paper-folder is: How do we know which sequence to fold in order to produce a particular regular star polygon with the FAT-algorithm? Let us demonstrate with an example, which is sufficiently general that it will apply to any regular star $\{\frac{b}{a}\}$ -gon where a, b are mutually prime odd integers with $a < (\frac{b}{2})$.

Suppose we desperately want to construct a regular star $\{\frac{11}{3}\}$ -gon. Then of course, $b = 11$, $a = 3$, and it is clear that if we begin with an angle of $(\frac{3}{11})\pi$ at A_0 and adhere to the Folding Rules, we will obtain a strip of tape with the angles and creases (indicated by dotted lines) shown in Figure 2.9.

We could denote this folding process as $D^1 U^3 D^1 U^1 D^3 U^1$, interpreted in the obvious way on the tape – that is, the first exponent “1” refers to the one bisection (producing a line in a downward direction) at the vertices A_{10n} (for $n = 0, 1, 2, \dots$) on the top of the tape; similarly, the “3” refers to the 3

bisections (producing creases in an upward direction) made at the bottom of the tape through the vertices A_{10n+1} ; etc. However, since the folding process is *duplicated* halfway through, we can abbreviate the notation and write simply $\{1, 3, 1\}$, with the understanding that we alternate folding from the top and bottom of the tape as described, with the number of bisections at each vertex running, in order, through the values $1, 3, 1, \dots$. We call this a *primary folding procedure of period 3*.

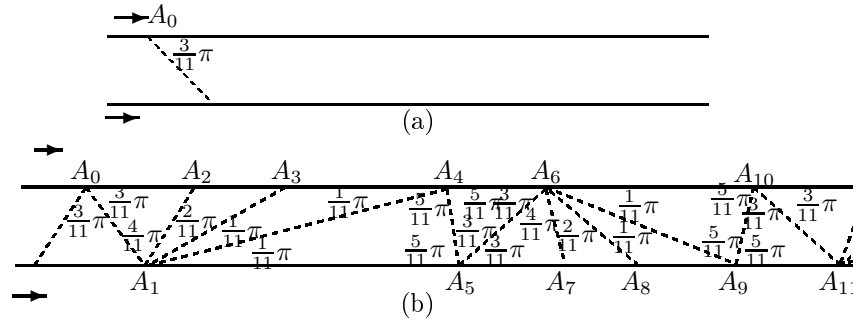


Figure 2.9

The proof of convergence for this folding procedure is similar to that for the primary folding procedure of period 2. We leave the details to the reader. Let us explore here what we can do with this $\{1, 3, 1\}$ -tape. First, note that starting with the putative angle $(\frac{3}{11})\pi$ at the top of the tape, we produce a putative angle of $(\frac{5}{11})\pi$ at the bottom of the tape, then a putative angle $(\frac{1}{11})\pi$ at the top of the tape, then a putative angle of $(\frac{3}{11})\pi$ at the *bottom* of the tape, and so on. Hence we see that we could use this tape to fold a star $\{\frac{11}{3}\}$ -gon, a convex 11-gon, and a star $\{\frac{11}{5}\}$ -gon. More still is true; for, as we see, if there are crease lines enabling us to fold a star $\{\frac{11}{a}\}$ -gon, there will be crease lines enabling us to fold star $\{\frac{11}{2^k a}\}$ -gons, where $k \geq 0$ takes all values such that $2^{k+1}a < 11$.

These features, as described for $b = 11$, apply for any odd number b . However, this tape has a special symmetry as a consequence of its *odd* period; namely that if it is “flipped” about the horizontal line halfway between its parallel edges, the result is a translate of the original tape. As a practical matter, this special symmetry of the tape means that we can use either the top edge or the bottom edge of the tape to construct our polygons. On tapes with an *even* period, the top edge and the bottom edge of the tape are not translates of each other (after the horizontal flip), which simply means that care must be taken in choosing the edge of the tape used to construct a specific polygon. Figures 2.10(a,b) show the completed $\{\frac{11}{3}\}$, $\{\frac{11}{4}\}$ -gons, respectively. (How would you guess we prepared these illustrations?!)

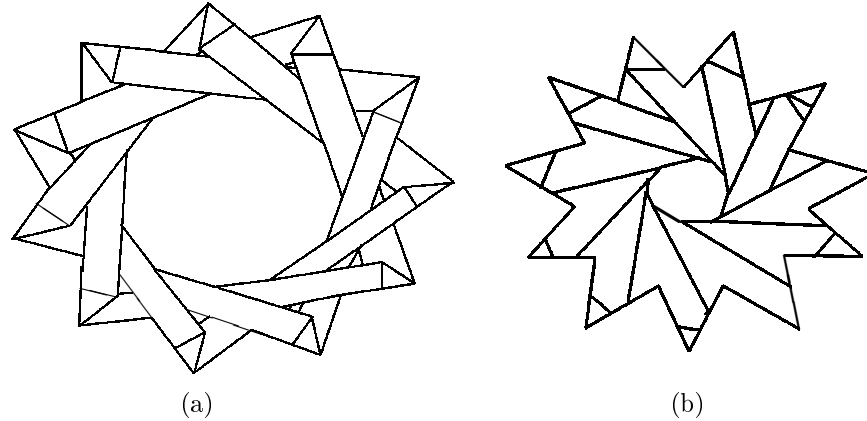


Figure 2.10

Now, to set the scene for the number theory, let us look at the patterns in the arithmetic of the computations when $a = 3$ and $b = 11$. Referring to Figure 2.9, we observe that

the angle to the right of A_n where $n =$	is of the form $(a_i/11)\pi$ where $a_i =$	and the number of bisections at $A_n =$
0	3	3
1	1	1
4	5	1
5	3	3
6	1	1
9	5	1

We could write this in shorthand form as follows:

$$(b=)11 \left| \begin{array}{ccc} (a=) & 3 & 1 & 5 \\ & 3 & 1 & 1 \end{array} \right| \quad (2.7)$$

Observe that had we started with the putative angle of $\frac{\pi}{11}$, then the *symbol* (2.7) would have taken the form

$$(b=)11 \left| \begin{array}{ccc} (a=) & 1 & 5 & 3 \\ & 1 & 1 & 3 \end{array} \right| \quad (2.7')$$

In fact, it should be clear that we can *start anywhere* (with $a = 1, 3$, or 5), and the resulting symbol, analogous to (2.7), will be the cyclic permutation of the interior that places our choice of a in the first position along the top row.

The generalization of this symbol appears at the beginning of Section 4. But before leaving this section, let us comment that given any two odd numbers a and b , with $a < (\frac{b}{2})$, there is always a completely determined unique symbol like the one above (we do not need a, b relatively prime). Appropriately interpreted, we can use this symbol to read off the folding procedure that produces the angle of $(\frac{a}{b})\pi$ along the top edge of the tape, so that a symbol such as (2.7) encodes

the folding procedure for producing a regular star $\{\frac{b}{a}\}$ -gon, and also tells us what other star polygons we can obtain from the same tape (of course, for each symbol, a diagram similar to Figure 2.9 can be drawn to illustrate the relative positions of the angles $(\frac{a}{b})\pi$).

The answer to question 4 is that the folding process described above is the *most efficient* one possible. That is, there could not be any folding procedure of this type that would produce the required star polygons with fewer folds. It is also optimal from the point of view of “difficulty of execution”, for it keeps the number of bisections at each vertex to a minimum.

These last comments are explained as follows. If a folding procedure $\{k_1, k_2, \dots, k_r\}$ produces an angle $(\frac{a}{b})\pi$, then (see (4.3), (4.4)) $b \mid 2^k \pm 1$, where $k = \sum_{i=1}^r k_i$. If we adopt the procedures described in this section, we will have a procedure $\{\ell_1, \ell_2, \dots, \ell_s\}$, such that $\ell = \sum_{j=1}^s \ell_j$ is the *smallest* number m such that $b \mid 2^m \pm 1$; that is, ℓ is the *quasi-order* of 2 mod b . In fact, $\ell \mid k$ and r will be a multiple of s . See Section 4 for details.

3 The Number Theory Of Period 2 Folding

We have seen that the $\{D^m U^n\}$ -procedure, or more briefly the (m, n) -procedure, enables us to fold star $\{\frac{b}{a}\}$ -gons, where $b = 2^{m+n} - 1$, $a = 2^n - 1$. We will generalize to the extent of replacing the base 2 by an arbitrary positive integer $t \geq 2$, so that we are concerned with rational numbers

$$\frac{t^u - 1}{t^v - 1}, \quad u > v \geq 1 \quad (3.1)$$

Our first result establishes how to reduce such fractions (3.1).

Theorem 3.1: Let $d = \gcd(u, v)$. Then $\gcd(t^u - 1, t^v - 1) = t^d - 1$.

Proof: Let us use base t , and let $\iota = t - 1$. Then

$$\begin{aligned} t^u - 1 &= \iota \iota \dots \iota & (u \text{ times}) \\ t^v - 1 &= \iota \iota \dots \iota & (v \text{ times}) \end{aligned}$$

It is now obvious that if we divide $(t^u - 1)$ by $(t^v - 1)$, the remainder is $(t^r - 1)$, where r is the remainder when we divide u by v . Thus, an application of the Euclidean algorithm establishes the theorem.

Remarks:

- (i) This theorem is propaganda for the Euclidean algorithm. We defy the reader to prove it by factoring $(t^u - 1)$, $(t^v - 1)$ into prime factors.
- (ii) The statement becomes false if we replace $\gcd$ by lcm . Thus if $\ell = \text{lcm}(u, v)$, then it is obvious that $\text{lcm}(t^u - 1, t^v - 1)$ divides $(t^\ell - 1)$, but it may divide $(t^\ell - 1)$ properly. For example, $\text{lcm}(2^2 - 1, 2^3 - 1) = 21$, and $21 \mid (2^6 - 1) = 63$.

Corollary 3.2: $\frac{(t^u - 1)}{(t^v - 1)}$ is an integer if and only if $v \mid u$.

We call such integers *t-folding numbers*, and write F_t for the set of such numbers. It is sometimes convenient to include 1 as a *degenerate t-folding number*; we write $\bar{F}_t$ for $F_t \cup \{1\}$.

Corollary 3.3: The rational number (3.1) reduces to the irreducible fraction $\frac{\beta}{\alpha}$, where $\alpha \in \bar{F}_t$, $\beta \in F_t$, and

$$\beta = \frac{t^u - 1}{t^d - 1}, \quad \alpha = \frac{t^v - 1}{t^d - 1}, \quad d = \gcd(u, v) \quad (3.2)$$

Let us change notation, setting $d = x$, $u = xy$, $v = xy'$. Thus

$$\beta = \frac{t^{xy} - 1}{t^x - 1}, \quad x \geq 1, \quad y \geq 2 \quad (3.3)$$

Then the question naturally arises of how we should express the possible values of α in terms of x and y . To answer this, let us write β in base t , so that

$$\beta = 10 \dots 0 \ 10 \dots 0 \ \dots \ 10 \dots 01 \quad (3.4)$$

where the *repeating* part $10 \dots 0$ consists of 1 followed by $(x-1)$ zeros, and there are y 1s on the right-hand side of equation (3.4). Then α is a (*right*) *section* of β , meaning that the t -numeral for α has the same repeating part but fewer 1s. In fact, it has y' 1s, because

$$\alpha = \frac{t^{xy'} - 1}{t^x - 1} \quad (3.5)$$

Since $\gcd(xy, xy') = x$, it follows that y' is restricted to those integers $< y$ which are prime to y . Thus we obtain $\phi(y)$ possible values of α , which correspond to the *relatively prime sections* of β .

Example: Consider the 2-folding number 21. Now, $21 = \frac{2^6-1}{(2^2-1)}$, so that $x = 2$, $y = 3$. Thus the possible values of α are given by (3.5) with $t = 2$, $x = 2$, and $y' = 1$ or 2 . Of course, $y' = 1$ gives $\alpha = 1$; and $y' = 2$ gives $\alpha = \frac{(2^4-1)}{(2^2-1)} = 5$. Thus our period 2 folding procedure enables us to fold the convex $2l$ -gon or the star $\{\frac{2l}{5}\}$ -gon. Obviously, the convex $2l$ -gon is obtained by the $(4, 2)$ -procedure, while the star $\{\frac{2l}{5}\}$ -gon is obtained by the $(2, 4)$ -procedure. (Alternatively, we can say that it is obtained by the $(4, 2)$ -procedure, but using the FAT-algorithm on the *bottom* edge of the tape.)

We now prove two basic theorems on t -folding numbers. For β of the form (3.3), the base t representation (3.4) shows that x, y are determined by β . We write $\tau(\beta) = xy$.

Theorem 3.4: Let $a > 1$ be an integer prime to t . Then for any x , there exists β of the form (3.3) with $a \mid \beta$. Moreover, there exists a positive integer $h(a)$, called the *height* of a , such that

- (i) there exists a $\beta_0 \in F_t$, with $a \mid \beta_0$ and $\tau(\beta_0) = h(a)$;
- (ii) if $\beta \in F_t$ and $a \mid \beta$, then $h(a) \mid \tau(\beta)$.

Proof: Now t is prime to a and to $t^x - 1$, so that t is prime to $a(t^x - 1)$. Let z be the order of $t \pmod{a(t^x - 1)}$. Then $t^z \equiv 1 \pmod{a(t^x - 1)}$. Since $t^x - 1 \mid t^z - 1$, it follows by Corollary 3.2 that $x \mid z$, so we may write $z = xy$. Also $y \geq 2$, since $a > 1$. Thus $\frac{(t^{xy}-1)}{(t^x-1)} \in F$, and $a \mid \frac{(t^{xy}-1)}{(t^x-1)}$. We use the above argument with $x = 1$ to obtain $\beta_0 = \frac{(t^y-1)}{(t-1)}$, and set $y_0 = h(a)$, establishing (i). Now, if $a \mid \frac{(t^{xy}-1)}{(t^x-1)}$, then $a \mid \frac{(t^{xy}-1)}{(t-1)}$, so that $t^{xy} \equiv 1 \pmod{a(t-1)}$. But $h(a)$ is the order of $t \pmod{a(t-1)}$, so that $h(a) \mid xy = \tau(\beta)$, establishing (ii).

Theorem 3.5: Let $a > 1$ be an integer prime to t , and let β be the smallest t -folding number such that $a \mid \beta$. Then $h(a) = \tau(\beta)$.

Proof: Let $\beta = \frac{(t^{xy}-1)}{(t^x-1)}$ so that $\tau(\beta) = xy$, and let $\beta_0 = \frac{(t^{h(a)}-1)}{(t-1)}$ as in the proof of Theorem 3.4. We show that if $h(a) \neq \tau(\beta)$, then $\beta > \beta_0$, contradicting the minimality of β . Now $h(a) \mid \tau(\beta)$, so that if $h(a) \neq \tau(\beta)$, then $h(a) \leq (\frac{1}{2}) \tau(\beta)$. Thus, since $y \geq 2$, we have

$$\beta > \frac{t^{xy}}{t^x} = t^{x(y-1)} \geq t^{xy/2} \geq t^{h(a)} > \beta_0$$

Remarks:

- (i) Theorem 3.5 provides us with an algorithm for finding the smallest β such that $a \mid \beta$. We first compute $h(a)$, and then test the t -folding numbers

$$\frac{t^{h(a)}-1}{t^{x_1}-1}, \frac{t^{h(a)}-1}{t^{x_2}-1}, \dots$$

as $x_1, x_2, \dots$ descend through the proper divisors of $h(a)$, to find the first of the sequence which is divisible by a . Notice that if a is prime, this must be $\frac{(t^{h(a)}-1)}{(t^{x_1}-1)}$. For

$$a \mid \frac{t^{h(a)}-1}{t^{x_1}-1} \cdot \frac{t^{x_1}-1}{t-1}, \text{ but } a \nmid \frac{t^{x_1}-1}{t-1}$$

the latter by the definition of $h(a)$. For example, if $a = 7$ and $t = 3$, then $h(a) = 6$, so the minimum β is $\frac{(3^6-1)}{(3^3-1)} = 28$.

- (ii) Theorem 3.5 is very useful to the paper-folder. For it affirms that in choosing the smallest folding number β having a given odd number a as a factor, we automatically involve ourselves in the smallest number of fold lines – recall that with $t = 2$, $\tau(\beta) = xy = m + n$.

Our basic theorems have the following surprising consequence.

Corollary 3.6: If for some t ,

$$\frac{t^{xy}-1}{t^x-1} \mid \frac{t^{x'y'}-1}{t^{x'}-1}$$

with $x, x' \geq 1$, $y, y' \geq 2$, then $xy \mid x'y'$.

Proof: Set $a = \frac{(t^{xy}-1)}{(t^x-1)}$. Since $a \in F_t$, it is obviously the smallest $\beta \in F_t$ such that $a \mid \beta$. Thus by Theorem 3.5, $h(a) = \tau(a) = xy$. But by Theorem 3.4(ii), $h(a) \mid \tau(\beta') = x'y'$, where $\beta' = \frac{(t^{x'y'}-1)}{(t^{x'}-1)}$.

This result is the natural starting point for a systematic study of divisibility questions. We consider t -folding numbers $\beta_t \mid \beta'_t$ as in Corollary 3.6, and ask: For what values of t do we have $\beta_t \mid \beta'_t$? Corollary 3.6 tells us that we may as well assume that $xy \mid x'y'$; otherwise, the divisibility condition holds for *no* values of t . Thus we assume $xy \mid x'y'$. We now give a complete answer to our question, which we enunciate as a Divisibility Theorem.

Set $d = \gcd(xy, x')$, $h = \gcd(x, x') = \gcd(x, d)$. Now $xy = ad$, $x' = bd$, where $\gcd(a, b) = 1$. Thus $ad \mid bdy'$, so $a \mid by'$, and hence $a \mid y'$. We can now formulate the promised answer.

Theorem 3.7 (Divisibility Theorem): Let $\beta = \frac{(t^{xy}-1)}{(t^x-1)}$, $\beta' = \frac{(t^{x'y'}-1)}{(t^{x'}-1)}$ be t -folding numbers. Then, assuming $xy \mid x'y'$,

$$\beta_t \mid \beta'_t \iff \frac{t^d - 1}{t^h - 1} \mid \frac{y'}{a}$$

We base our proof on 3 easy lemmas. For the first, we allow ourselves to write $\left(\frac{m}{n}\right) \mid k$ if $m \mid nk$, even if $n \nmid m$.

Lemma 3.8: $\left(\frac{m}{n}\right) \mid k \iff \left(\frac{m}{g}\right) \mid k$, where $g = \gcd(m, n)$.

Proof: If $m \mid gk$, then $m \mid nk$. Conversely, let $m = m'g$, $n = n'g$. Then if $m \mid nk$, $m' \mid n'k$, $m' \mid k$ or $\left(\frac{m}{g}\right) \mid k$.

Lemma 3.9: If $g = \gcd(u, v)$, then $\text{lcm}(gu, gv) = uv$.

Proof: Let $\ell = \text{lcm}(u, v)$. Then $g\ell = uv$. But $\text{lcm}(gu, gv) = g\text{lcm}(u, v) = g\ell$.

Lemma 3.10: Let $d \mid q$. Then

$$\frac{t^d - 1}{t^x - 1} \mid \frac{t^{qr} - 1}{t^q - 1} \iff \frac{t^d - 1}{t^x - 1} \mid r$$

Proof: Note first that by Lemma 3.8 and Theorem 3.1, we may replace x by h in this statement. To prove it, we set $t^d = s$, $q = dk$, and compute the remainder on dividing the polynomial $\frac{(s^{kr}-1)}{(s^k-1)}$ by $s-1$. By L'Hopital's Rule, this remainder is

$$\lim_{s \rightarrow 1} \frac{s^{kr} - 1}{s^k - 1} = r$$

Thus $\left(\frac{(t^{qr}-1)}{(t^q-1)}\right) = Q(t^d)(t^d - 1) + r$, where Q is a polynomial with integer coefficients.

It follows that

$$\frac{t^{qr} - 1}{t^q - 1} = (t^h - 1)Q(t^d)\frac{t^d - 1}{t^h - 1} + r$$

from which the lemma immediately follows.

Proof of Theorem 3.7: Set $u = t^{xy} - 1$, $v = t^{x'} - 1$. Then $g = \gcd(u, v) = t^d - 1$, by Theorem 3.1. If $c = (t^x - 1)(t^{x'y'} - 1)$, then

$$\beta_t \mid \beta'_t \iff uv \mid c \iff (t^d - 1)(t^{xy} - 1) \mid c \text{ and } (t^d - 1)(t^{x'} - 1) \mid c, \text{ by Lemma 3.9}$$

$$\begin{aligned} &\iff \frac{t^d - 1}{t^x - 1} \mid \frac{t^{x'y'} - 1}{t^{xy} - 1} \quad \text{and} \quad \frac{t^d - 1}{t^x - 1} \mid \frac{t^{x'y'} - 1}{t^{x'} - 1} \\ &\iff \frac{t^d - 1}{t^x - 1} \mid \frac{x'y'}{xy} \quad \text{and} \quad \frac{t^d - 1}{t^x - 1} \mid y', \text{ by Lemma 3.10} \\ &\iff \frac{t^d - 1}{t^h - 1} \mid \frac{x'y}{xy} \quad \text{and} \quad \frac{t^d - 1}{t^h - 1} \mid y', \text{ by Thm 3.1 / Lem. 3.8} \\ &\iff \frac{t^d - 1}{t^h - 1} \mid \gcd\left(\frac{x'y}{xy}, y'\right) \end{aligned}$$

Now $x'y' = bdy'$, $xy = ad$, so $\frac{x'y'}{xy} = \frac{by'}{a}$. Thus $\gcd\left(\left(\frac{x'y'}{xy}\right), y'\right) = \gcd\left(b\left(\frac{y'}{a}\right), a\left(\frac{y'}{a}\right)\right) = \frac{y'}{a}$, since a, b are mutually prime. Thus, finally,

$$\beta_t \mid \beta'_t \iff \frac{t^d - 1}{t^h - 1} \mid \frac{y'}{a}$$

We draw some consequences from this theorem; we preserve all the hypotheses and notation of the theorem.

Corollary 3.11: Either $d \mid x$ and $\beta_t \mid \beta'_t$ for all t , or $d \nmid x$ and $\beta_t \mid \beta'_t$ for only finitely many t .

Corollary 3.12: $\left[\frac{(t^{xy}-1)}{(t^x-1)} \right] \mid \left[\frac{(t^{x'y'}-1)}{(t^{x'}-1)} \right] \iff \left[\frac{(t^{xy}-1)}{(t^h-1)} \right] \mid \left[\frac{(t^{x'y'}-1)}{(t^x-1)} \right].$

We also have the following interesting result.

Theorem 3.13: The following conditions are equivalent:

- (i) $\beta_t \mid \beta'_t$ for all t ;
- (ii) $\beta_t \mid \beta'_t$ for infinitely many t ;
- (iii) $\beta_t \mid \beta'_t$ as polynomials in $Z[t]$;
- (iv) $xy \mid x'y'$ and $d \mid x$.

Proof: We already know the equivalence of (i), (ii), and (iv). Since plainly (iii) $\Rightarrow$ (i), it remains to show that (i) $\Rightarrow$ (iii). This follows from the following lemma.

Lemma 3.14: Let $f(t), g(t)$ be polynomials in $Z[t]$ with $g(t)$ monic, and suppose that $g(m) \mid f(m)$ for infinitely many integers m . Then $g(t) \mid f(t)$.

Proof: Since $g(t)$ is monic, we may execute the division algorithm in $Z[t]$, obtaining

$$f(t) = q(t)g(t) + r(t), \text{ where } r(t) = 0 \text{ or } \deg r < \deg g$$

We will obtain a contradiction from assuming $r(t) \neq 0$. Now

$$g(m) \mid f(m) \iff g(m) \mid r(m)$$

Moreover, since $r(t)$ or $g(t)$ vanishes for only finitely many values of t , we may suppose that $g(m) \mid r(m)$ for infinitely many m , for none of which either $r(t)$ or $g(t)$ vanishes. We consider the function $\frac{r(t)}{g(t)}$. Then $\left| \frac{r(t)}{g(t)} \right| \geq 1$ for infinitely many integral values of t . On the other hand, since $\deg r < \deg g$, $\frac{r(t)}{g(t)} \rightarrow 0$ as $t \rightarrow \infty$ and as $t \rightarrow -\infty$. Thus $\left| \frac{r(t)}{g(t)} \right| \leq \frac{1}{2}$ for $|t|$ sufficiently large. This is the contradiction we seek.

Examples:

- (1) Let $\beta_t = \left[\frac{(t^3-1)}{(t-1)} \right]$, $\beta'_t = \left[\frac{(t^6-1)}{(t^2-1)} \right]$. Here $d = 1$, $x = 1$, so we are in the situation covered by Theorem 3.13. Indeed,

$$\begin{aligned} \beta_t &= t^2 + t + 1, \\ \beta'_t &= t^4 + t^2 + 1, \text{ and} \\ t^4 + t^2 + 1 &= (t^2 + t + 1)(t^2 - t + 1) \end{aligned}$$

Writing in base t , we have $\beta_t = 111$, $\beta'_t = 10101$, and we know that

$$111 \mid 10101, \text{ for all } t$$

With $t = 2$, we have $7 \mid 21$; with $t = 3$, we have $13 \nmid 91$; with $t = 4$, we have $21 \mid 273$; $\dots$.

- (2) Let $\beta_t = \left[\frac{(t^2-1)}{(t-1)} \right]$, $\beta'_t = \left[\frac{(t^6-1)}{(t^2-1)} \right]$. Here $d = 2$, $x = 1$, so $h = 1$; also, $a = b = 1$, $y' = 3$, so

$$\beta_t \mid \beta'_t \iff \frac{t^2-1}{t-1} \mid 3 \iff (t+1) \mid 3$$

Since $t \geq 2$ is an integer, this means that $\beta_t \mid \beta'_t \iff t = 2$. Thus the assertion $11 \mid 10101$ is only true in base 2. There it asserts that $3 \mid 21$; we verify that $4 \nmid 91$, $5 \nmid 273$, $6 \nmid 651$, $\dots$.

These two examples might be (lightheartedly!) summed up by saying that $7 \mid 21$ always, while $3 \mid 21$ almost never!

- (3) Let $\beta_t = \left[\frac{(t^{24}-1)}{(t^6-1)} \right]$, $\beta'_t = \left[\frac{(t^{60y}-1)}{(t^{60}-1)} \right]$. Since we need $24 \mid 60y'$, we must have y' even, if $\beta_t \mid \beta'_t$ for any t . Now $d = 12$, $h = 6$, so $a = 2$ and

$$\beta_t \mid \beta'_t \iff \frac{t^{12}-1}{t^6-1} \mid \frac{y'}{2}$$

We thus see that $\beta_t \nmid \beta'_t$ for all t , if $y' < 130$. If $y' = 130$, then $\beta_t \mid \beta'_t$ just for $t = 2$. The smallest y' such that $\beta_3 \mid \beta'_3$ is given by $y' = 1460$. Arguing the other way, we see that if $y' \leq 2 \cdot 10^6$, then the only possible values of t such that $\beta_t \mid \beta'_t$ are in the range $2 \leq t \leq 9$.

We close this section with a divisibility property, relating to t -folding numbers, of a very different kind.

Theorem 3.15: Let $\beta \in F_t$. Then

- (i) the order of $t \pmod{\beta}$ is $\tau(\beta)$,
- (ii) $\tau(\beta) + \Phi(\beta)$, where Φ is the Euler totient function.

Proof:

- (i) If $\beta = \left[\frac{(t^{xy}-1)}{(t^x-1)} \right]$, then $xy = \tau(\beta)$ and $t^{xy} \equiv 1 \pmod{\beta}$. Thus if h is the order of t modulo β , then $h \mid xy$. Suppose $h \sim xy$. Then $h \leq \frac{1}{2}xy$, so (compare the proof of Theorem 3.5):

$$\beta_t > \frac{t^{xy}}{t^x} = t^{x(y-1)} \geq t^{xy/2} \geq t^h > t^h - 1$$

so it is not possible that $t^h \equiv 1 \pmod{\beta}$. Thus $h = xy$.

- (ii) Since $\Phi(\beta)$ is the order of the multiplicative group of residues prime to β , it follows that $\tau^{\Phi(\beta)} = 1 \pmod{\beta}$. Thus from (i), $\tau(\beta) \mid \Phi(\beta)$.

It is again interesting to remark that, while $\Phi(\beta)$ certainly depends on t , $\tau(\beta)$ does not. Thus we may re-express Theorem 3.15 (ii) as follows. Let q be a fixed positive integer. Consider all the integers β of the form $\frac{(t^q-1)}{(t^r-1)}$, where t is an arbitrary integer ≥ 2 , and r is an arbitrary proper factor of q . Then $q \mid \Phi(\beta)$ for any such integer β .

Finally, we note that an even stronger result than Theorem 3.15 is available. For it follows easily from Theorem 4.10 below, that if $\beta = \frac{(t^{xy}-1)}{(t^x-1)}$, so that $\tau(\beta) = xy$, then $\tau(\beta)$ is the *quasi-order* of $t \pmod{\beta}$ unless $y = 2$, when the quasi-order is $x = \left(\frac{1}{2}\right)\tau(\beta)$ (see Theorem 4.10 for the precise definition of quasi-order). Since we know that $t^{\tau(\beta)} \equiv 1 \pmod{\beta}$, this is a finer result than Theorem 3.15 (i). As to Theorem 3.15 (ii), we can improve this too by replacing $\Phi(\beta)$ by the *exponent* $\psi(\beta)$, of the multiplicative group of residues prime to β . For example, if $\beta = 91 = \frac{(3^6-1)}{(3^2-1)}$, then $\tau(\beta) = 6$, $\Phi(\beta) = 72$, $\psi(\beta) = 12$.

4 The Number Theory Of The General Folding Procedure

Let

$$b \begin{vmatrix} a_1 & a_2 & \cdots & a_r \\ k_1 & k_2 & \cdots & k_r \end{vmatrix} \quad (4.1)$$

be a (2-)symbol as presented in Section 2, where only special cases (e.g. (2.7)) were described explicitly. Thus b, a_i ($a_1 = a$) are odd, $a_1 < \frac{b}{2}$, and

$$b = a_i + 2^{k_i} a_{i+1}, \quad i = 1, 2, \dots, r, \quad a_{r+1} = a_1 \quad (4.2)$$

At this stage, we do not assume that $\gcd(b, a) = 1$, nor do we assume that the list $a_1, a_2, \dots, a_r$ is without repeats. Indeed, if $\gcd(b, a) = 1$, we say that the symbol (4.1) is *reduced*; and if there are no repeats, we say that the symbol (4.1) is *contracted*. Thus the folding instructions for regular star polygons of Section 2 are encoded in reduced contracted symbols.

We now study the number theory embodied in the symbols (4.1). Our first observation is a trivial consequence of (4.2).

Lemma 4.1:

- (i) Given the symbol (4.1), then $\gcd(b, a_i)$ is independent of i and $a_i < \frac{b}{2}$, $1 \leq i \leq r$.
- (ii) (4.1) is a symbol if and only if

$$qb \begin{vmatrix} qa_1 & qa_2 & \cdots & qa_r \\ k_1 & k_2 & \cdots & k_r \end{vmatrix}$$

is a symbol, where q is an arbitrary odd number. (This symbol is said to arise by *inflating* (4.1).)

If we regard (4.2) as a set of r simultaneous linear equations in the “unknowns” $a_1, a_2, \dots, a_r$, then their solution is given by

$$Ba_i = bA_i, \quad i = 1, 2, \dots, r \quad (4.3)$$

where

$$B = 2^k - (-1)^r, \quad k = \sum k_i \quad (4.4)$$

and

$$A_i = 2^{k-k_{i-1}} - 2^{k-k_{i-1}-k_{i-2}} + \cdots + (-1)^{r_2^{k_i}} - (-1)^r, \quad i = 1, 2, \dots, r \quad (4.5)$$

Here, and subsequently, we find it convenient to think of the suffices i as running over the residue classes $\mod r$, so that $k_0 = k_r$, etc. We now announce a portmanteau lemma.

Lemma 4.2:

- (i) The symbol (4.1) is determined up to contradiction by b and a_1 .
- (ii) The symbol (4.1) is determined by b and $k_1, k_2, \dots, k_r$.
- (iii)

$$B \begin{vmatrix} A_1 & A_2 & \cdots & A_r \\ k_1 & k_2 & \cdots & k_r \end{vmatrix} \quad (4.6)$$

is a symbol.

- (iv) If (4.1) is reduced, then it is obtained by reducing (4.6).
- (v) A_i is independent of k_{i-1} .

Proof: (i) follows from the definitions.

(ii) is clear from (4.3, 4.4, 4.5).

(iii) follows by observing that

$$b' \begin{vmatrix} a'_1 & a'_2 & \cdots & a'_r \\ k_1 & k_2 & \cdots & k_r \end{vmatrix}$$

is a symbol provided that b', a'_i are odd integers satisfying $a'_i < \left(\frac{b'}{2}\right)$, and (4.3) with b', a'_i replacing b, a_i . But clearly $A_i < (B/2)$, and the equations $Ba'_i = b'A_i$ are obviously satisfied by $b' = B, a'_i = A_i$.

- (iv) If $\gcd(b, a_i) = 1$, then from (4.3), there exists a q (odd) such that $B = qb, A_i = qa_i$.
- (v) is a consequence of (4.5).

We call (4.6) a *canonical* symbol, or, more precisely, the *canonical symbol associated with* $(k_1, k_2, \dots, k_r)$. As we see, every symbol is obtained by reducing a canonical symbol and (perhaps) then inflating the resulting symbol. Thus while symbols do in a sense generalize the notion of a fraction, we see that there are two kinds of standard symbol: the reduced contracted symbols, and the canonical symbols.

Of course, canonical symbols need not be reduced. The following lemma deals with the question of whether they are contracted.

Lemma 4.3:

- (i) If the sequence $k_1, k_2, \dots, k_r$ is obtained by repeating a subsequence $k_1, k_2, \dots, k_s$, then the sequence $a_1, a_2, \dots, a_r$ is obtained by repeating the subsequence $a_1, a_2, \dots, a_s$.
- (ii) The canonical symbol (4.6) is contracted if and only if the symbol (4.1) is contracted.

Proof:

- (i) We suppose that there exists an s such that $s \mid r, s \neq r$, and $k_{s+i} = k_i$ for all i . It is then plain from (4.5) that $A_{s+i} = A_i$ for all i , so that by (4.3), $a_{s+i} = a_i$ for all i .

- (ii) Using (i), we observe that the symbol (4.1) (and the symbol (4.6)) involves a repeat if and only if the sequence $k_1, k_2, \dots, k_r$ involves a repeat.

Remarks:

- (i) When we talk of the symbol (4.1) *repeating*, we require only that for some s, t with $1 \leq s, t \leq r$, and $s \neq t$, $a_s = a_t$. It is easy to see that if this happens, then in fact a_1 must repeat, say with $a_n = a_1$, and then the entire symbol repeats with “period” $(n - 1)$. However, in Lemma 4.3, we do not merely require that some k_i is repeated, but that the sequence repeats in the sense described.
- (ii) We particularly note that Lemma 4.2 (iii),(iv) answers our question as to what regular star $\left\{\frac{b}{a}\right\}$ -gons are obtained by the folding instruction $(k_1, k_2, \dots, k_r)$, which is of course assumed to be non-repeating. Thus we calculate $B, A_1, \dots, A_r$ from (4.4), (4.5) and then reduce by dividing by $d = \gcd(B, A_i)$, which is known to be independent of i . If $B = bd$, $A_i = a_i d$, then we can construct by the given folding instruction, precisely the $\left\{\frac{b}{a_1}\right\}$ -gons. This, of course, assumes we are prepared to use the FAT-algorithm on either the top or the bottom of the tape.

We now move towards our principal number-theoretical result. (All integers referred to are positive.)

Lemma 4.4: Let b be odd ≥ 3 , and let S be the set of odd integers a with $a < \frac{b}{2}$. Then the function ψ on S , given by $\psi(a) = a'$, where $b = a + 2^k a'$, a' odd, is a permutation of S .

Proof: Since $k \geq 1$, it is obvious that $a' < \frac{b}{2}$. We show that ψ is a permutation of S by defining an inverse φ . Thus, given $a' \in S$, let k be maximal with $2^k a' < b$. Then $k \geq 1$, so that if $a = b - 2^k a'$, then a is odd and $a < \frac{b}{2}$. Of course, we define $\varphi(a') = a$.

This lemma shows that, as claimed, a symbol always exists for given odd numbers b, a with $a < \frac{b}{2}$; and it is, of course, unique up to contraction.

Theorem 4.5 (Quasi-Order Theorem in base 2): Let (4.1) be a reduced contracted symbol, and let $k = \Sigma k_i$. Then $2^k \equiv (-1)^r \pmod{b}$. Indeed, k is the quasi-order of $2 \pmod{b}$; that is, the smallest positive integer n such that $2^n \equiv \pm 1 \pmod{b}$.

Proof: It is obvious from (4.3), since $\gcd(b, a_i) = 1$, that $b \mid 2^k - (-1)^r$. To prove that k is the quasi-order, we effectively employ the φ function introduced in the proof of Lemma 4.4. Thus consider the sequence of $(k + 1)$ positive integers $< \frac{b}{2}$ (see (4.2)).

$$a_r, 2a_r, \dots, 2^{k_{r-1}-1}a_r, a_{r-1}, 2a_{r-1}, \dots, a_{i+1}, 2a_{i+1}, \dots, 2^{k_i-1}a_{i+1}, \\ a_i, 2a_i, \dots, a_1, 2a_1, \dots, a^{k_r-1}a_1, a_r$$

Call this sequence $s_1, s_2, \dots, s_{k+1}$. Then plainly,

$$s_{j+1} \equiv \pm 2s_j \pmod{b}$$

the minus sign obtaining precisely when s_{j+1} is some a_i . (This provides an alternative proof that $2^k \equiv (-1)^r \pmod{b}$.)

Thus, if $2^\ell \equiv \pm 1 \pmod{b}$ for some positive $\ell < k$, we would have $s_{\ell+1} \equiv \pm a_r \pmod{b}$. We show that this is impossible. For if $s_{\ell+1} \equiv a_r \pmod{b}$, then since $s_{\ell+1}, a_r < \frac{b}{2}$, we have $s_{\ell+1} = a_r$. This is impossible if $s_{\ell+1} = 2^m a_i$, $m \geq 1$, since a_r is odd; but it is also impossible if $s_{\ell+1} = a_i$, since $i \not\mid r$ and the symbol is contracted. On the other hand, if $s_{\ell+1} \equiv -a_r \pmod{b}$, then $b \mid (s_{\ell+1} + a_r)$, which is impossible since $0 < s_{\ell+1} + a_r < b$. This proves the theorem.

There are many interesting consequences of this theorem. We give one of significance for paper-folding.

Corollary 4.6:

- (i) Let $k \geq 3$. Then every contracted reduced symbol (4.1) with $b = 2^k - 1$ is canonical.
- (ii) Let $k \geq 1$. Then every contracted reduced symbol (4.1) with $b = 2^k + 1$ is canonical.

Proof:

- (i) Obviously, the quasi-order of $2 \pmod{2^k - 1}$ is k if $k \geq 3$. Thus if $\gcd(2^k - 1, a_1) = 1$, the contracted reduced symbol we obtain, namely

$$2^k - 1 \left| \begin{array}{cccc} a_1 & a_2 & \cdots & a_r \\ k_1 & k_2 & \cdots & k_r \end{array} \right| \quad (4.7)$$

has by Theorem 4.5, $\Sigma k_i = k$ (and r even). By Lemma 4.2 (ii,iii), it follows that this symbol is canonical.

- (ii) is proved similarly.

Remark: We need $k \geq 3$ in (i), since $b \geq 3$, and if $k = 2$, then the quasi-order of $2 \pmod{2^k - 1}$ is 1, not 2.

Proposition 4.7: Let $k = \Sigma k_i$.

- (i) Let $k \geq 2$. Then if (4.7) is a symbol, there is a symbol

$$2^{k+\ell} - 1 \left| \begin{array}{ccccc} a_1 & a_2^{(\ell)} & \cdots & a_{r-1}^{(\ell)} & a_r^{(\ell)} \\ k_1 & k_2 & \cdots & k_{r-1} & k_r + \ell \end{array} \right| \quad (4.8-)$$

for some $a_2^{(\ell)}, \dots, a_r^{(\ell)}$, for any $\ell \geq 0$.

(ii) Let $k \geq 1$. Then if

$$2^k + 1 \left| \begin{array}{cccc} a_1 & a_2 & \cdots & a_r \\ k_1 & k_2 & \cdots & k_r \end{array} \right|$$

is a symbol, there is a symbol

$$2^{k+\ell} + 1 \left| \begin{array}{ccccc} a_1 & a_2^{(\ell)} & \cdots & a_{r-1}^{(\ell)} & a_r^{(\ell)} \\ k_1 & k_2 & \cdots & k_{r-1} & k_r + \ell \end{array} \right| \quad (4.8+)$$

for some $a_2^{(\ell)}, \dots, a_r^{(\ell)}$, for any $\ell \geq 0$.

Proof:

(i) If (4.7) is a symbol, then by (4.3),

$$(2^k - (-1)^r)a_1 = (2^k - 1)A_1 \quad (1)$$

The assumption that r is odd leads to the conclusion $2^k - 1 \mid a_1$ contradicting $a_1 < (\frac{1}{2})(2^k - 1)$. Thus r is even, $a_1 = A_1$, and (4.7) is canonical. Recalling Lemma 4.2 (v), it follows that the canonical symbol associated with $k_1, k_2, \dots, k_{r-1}, k_r + \ell$ has the same A_1 as the canonical symbol associated with $k_1, k_2, \dots, k_{r-1}, k_r$. This proves (i).

(ii) is proved similarly.

Corollary 4.8:

- (i) Let $\gcd(2^k - 1, a) = 1$, $k \geq 3$. If the regular $\left\{\frac{(2^k-1)}{a}\right\}$ -star polygon is folded by the procedure $(k_1, k_2, \dots, k_r)$, then the regular $\left\{\frac{(2^{k+\ell}-1)}{a}\right\}$ -star polygon is folded by the procedure $(k_1, k_2, \dots, k_{r-1}, k_r + \ell)$.
- (ii) Let $\gcd(2^k + 1, a) = 1$, $k \geq 1$. If the regular $\left\{\frac{(2^k+1)}{a}\right\}$ -star polygon is folded by the procedure $(k_1, k_2, \dots, k_r)$, then the regular $\left\{\frac{(2^{k+\ell}+1)}{a}\right\}$ -star polygon is folded by the procedure $(k_1, k_2, \dots, k_{r-1}, k_r + \ell)$.

Proof: This is an immediate consequence of Proposition 4.7 and Corollary 4.6.

Remark: For some ℓ , $\frac{(2^{k+\ell} \pm 1)}{a}$ may fail to be a reduced fraction. Moreover, it may happen (for at most one value of ℓ – see [5]) that the sequence $k_1, k_2, \dots, k_{r-1}, k_r + \ell$ repeats. In this latter very special case, we will certainly know from Theorem 4.5 that $\frac{(2^{k+\ell} \pm 1)}{a}$ is not reduced, and we will be able to contract the folding instructions.

We now consider how to generalize the notion of a 2-symbol to that of a t -symbol to obtain a generalization of Theorem 4.5. It turns out to be easier to generalize the φ -algorithm than the ψ -algorithm in Lemma 4.4.

In order to accommodate both the ψ -algorithm and the φ -algorithm, it is preferable to revise the format of a general 2-symbol, so that it appears as

$$b \left| \begin{array}{cccccc} a_1 & a_2 & \cdot & \cdot & \cdot & a_r & a_1 \\ & k_1 & & k_2 & \cdot & \cdot & \\ & & & & & & k_r \end{array} \right| \quad (4.9)$$

Then, if we are emphasizing the φ -algorithm, we would rewrite (4.9) as

$$b \left| \begin{array}{cccccc} c_1 & c_r & \cdot & \cdot & \cdot & c_2 & c_1 \\ & \ell_r & & & \ell_2 & & \ell_1 \end{array} \right| \quad (4.10)$$

so that $b = c_{i+1} + 2^{\ell_i} c_i$, and $\varphi(c_i) = c_{i+1}$, $i = 1, 2, \dots, r$ ($c_{r+1} = c_1$). Thus the data for the φ -algorithm with $t = 2$ are the following: we take b odd, and consider the set S of odd positive integers $c < \frac{b}{2}$. For such $c \in S$, let ℓ be maximal subject to $2^\ell c < b$, and define $c' = \varphi(c)$ by the rule

$$b = c' + 2^\ell c \quad (4.11)$$

To generalize from $t = 2$ to the case of a general positive integer $t \geq 2$, we find we should proceed as follows:

- (1) The condition that b is odd is generalized to the condition that b is *prime to t* .
- (2) The condition that c is odd is generalized to the condition that $t \nmid c$.
- (3) The condition $c < \frac{b}{2}$ is generalized to the condition $c < \frac{b}{t}$ (this was the most surprising).

Having generalized the set S , it remains to generalize the φ -algorithm. We first generalize the characterization of ℓ above.

- (4) The condition that ℓ is *maximal* subject to $2^\ell c < b$ is equivalent to the condition that ℓ is *minimal* subject to $2^\ell c > \frac{b}{2}$, and may therefore be generalized to the condition that ℓ is *minimal subject to $t^\ell c > \frac{b}{t}$* ; note that, as with $t = 2$, it follows that $\ell \geq 1$.

We are now ready to generalize φ . We observe that in (4.11), the multiple of b nearest to $2^\ell c$ is b itself; in fact, $b > 2^\ell c$, and $c' = \varphi(c)$ is the distance from $2^\ell c$ to the nearest multiple of b , namely to b itself. In general, we define $\varphi(c)$ to be the distance from $t^\ell c$ to the nearest multiple of b , where ℓ is defined as in (4) above. We immediately note that this definition makes sense since we cannot have $t^\ell c = \frac{nb}{2}$, for this would imply $b \mid 2c$, whereas $2c < b$. Thus if $\varphi(c) = c'$, then

$$qb + (-1)^\epsilon c' = t^\ell c, \quad \epsilon = 0 \text{ or } 1 \quad (4.12)$$

with $q \geq 1$. (Note that if $t = 2$, then $q = 1$ and $\epsilon = 1$.)

We remark that the minimality of ℓ implies that $t^\ell c < \frac{tb}{2}$, so that in (4.12), $q \leq \frac{t}{2}$ (and $q < \frac{t}{2}$ if $\epsilon = 0$). In fact, the restriction on q in (4.12) is precisely

$$\begin{cases} 1 \geq q \geq \frac{t-1}{2} & \text{if } t \text{ is odd} \\ 1 \geq q \geq \frac{t}{2} & \text{if } t \text{ is even and } \epsilon = 1 \\ 1 \geq q \geq \frac{t-2}{2} & \text{if } t \text{ is even and } \epsilon = 0 \end{cases} \quad (4.13)$$

The definition of φ makes it obvious that $0 < c' < \frac{b}{2}$. Since $q \leq \frac{t}{2}$, it follows that $t \nmid q$, so $t \nmid qb$, and hence from (4.12), $t \nmid c'$. Thus $c' \in S$, and φ is a function

$$\varphi : S \rightarrow S$$

Theorem 4.9: The function φ is a permutation of S .

Proof: First let t be odd, and consider for a given $c' \in S$, the set $Q = \{qb + (-l)^\epsilon c, 1 \leq q \leq \frac{(t-1)}{2}, \epsilon = 0, 1\}$. We claim that exactly one of the $(t-1)$ numbers in Q is divisible by t . Certainly, exactly one of the t numbers $sb + c', -\frac{(t-1)}{2} \leq s \leq \frac{(t-1)}{2}$, is divisible by t ; and this number cannot be given by $s = 0$, since $t \nmid c'$. If this number has s positive, set $s = q$, so that $t \mid (qb + c')$, $1 \leq q \leq \frac{(t-1)}{2}$; if it has s negative, set $s = -q$, so that $t \mid (qb - c')$, $1 \leq q \leq \frac{(t-1)}{2}$. Thus we have found q, ϵ satisfying $1 \leq q \leq \frac{(t-1)}{2}, \epsilon = 0, 1$, such that

$$t^\ell c = qb + (-1)^\epsilon c'$$

with $\ell \geq 1$ and $t \nmid c$. It remains to show that $t^{\ell-1}c < \frac{b}{2}$, since this guarantees both the minimality of ℓ and the fact that $c \in S$. But $t^\ell c < \left(\frac{(t-1)}{2}\right)b + \frac{b}{2} = \frac{tb}{2}$, establishing the required inequality. Thus, $c \in S$ and $\varphi(c) = c'$.

A small modification is required if t is even. We now consider the set

$$Q' = \left\{ qb + c', 1 \leq q \leq \frac{t}{2} - 1; \quad qb - c', 1 \leq q \leq \frac{t}{2} \right\}$$

Once again, we argue that exactly one of the $(t-1)$ numbers in Q' is divisible by t , and we choose it to get the equation

$$t^\ell c = qb + (-1)^\epsilon c'$$

where $\ell > 1$, $t \nmid c$, and $1 \leq q \leq \frac{t}{2}$ if $\epsilon = 1$, while $1 \leq q \leq \frac{t}{2} - 1$ if $\epsilon = 0$. We again argue that $t^{\ell-1}c < \frac{b}{2}$. For if $\epsilon = 0$, then $t^\ell c < \left(\frac{t}{2} - 1\right)b + \frac{b}{2} < \frac{tb}{2}$; and if $\epsilon = 1$, then $t^\ell c < \left(\frac{t}{2}\right)b$. This completes the proof of the theorem, for we have shown $\varphi : S \rightarrow S$ to be a surjective function from a finite set to itself, and hence a permutation.

Remarks:

- (i) The proof of the theorem implicitly contains the definition of the inverse algorithm $\psi : S \rightarrow S$. Notice that this ψ generalizes the algorithm ψ of Lemma 4.4, since with $t = 2$ we must take $\epsilon = 1$, $q = 1$, so that $\psi(a) = a'$, where $2^\ell a' = b - a$.
- (ii) It will be convenient to enlarge S to allow $a \leq \frac{b}{2}$; and when b is even, to define $\varphi(\frac{b}{2}) = \frac{b}{2}$, $\psi(\frac{b}{2}) = \frac{b}{2}$. Thus the choice for (4.12) which we make in this singular case is

$$\frac{t-1}{2}b + \frac{b}{2} = t\frac{b}{2}$$

which accords with the restriction $1 \leq q \leq \frac{(t-1)}{2}$ in the case that t is odd. We write $\bar{S}$ for $S \cup \{\frac{b}{2}\}$, for b even; and may write $\bar{S}$ for S if b is odd.

Our t -symbol is now more elaborate than in the case $t = 2$, and contains 4 rows, though only the second and third rows are relevant to the quasi-order theorem. Thus let b, t be mutually prime, and let $a_1 \in S$. We iterate the ψ -algorithm, getting

$$t^{k_i} a_{i+1} = q_i b + (-1)^{\epsilon_i} a_i, \quad i = 1, 2, \dots, r, \quad a_{r+1} = a_1 \quad (4.14)$$

where the positive integers q_i are restricted as in the proof of Theorem 4.9, and record (4.14) in the symbol (in its new format)

$$b \left| \begin{array}{cccccc} a_1 & a_2 & \cdot & \cdot & \cdot & a_r & a_1 \\ k_1 & k_2 & \cdot & \cdot & \cdot & k_r & \\ \epsilon_1 & \epsilon_2 & \cdot & \cdot & \cdot & \epsilon_r & \\ q_1 & q_2 & \cdot & \cdot & \cdot & q_r & \end{array} \right| t$$

$$k = \Sigma k_i, \quad \epsilon = \Sigma \epsilon_i \quad (4.15)$$

Note again that if $t = 2$, then all $\epsilon_i = 1$ and all $q_i = 1$, so that the last two rows of (4.15) may be omitted, and $\epsilon = r$.

We may clearly introduce the notions *contracted*, *reduced* for t -symbols. We will not copy all the earlier material on 2-symbols in this more general context, but will proceed immediately to enunciate the general quasi-order theorem.

Theorem 4.10: Let the symbol (4.15) be contracted and reduced. Then $t^k \equiv (-1)^\epsilon \pmod{b}$, and k is the quasi-order of $t \pmod{b}$.

Proof: We argue just as in the case $t = 2$ from the φ -algorithm. We leave the details to the reader.

Remark: By including the case $b = 2$, $a_1 = 1$, we make our quasi-order theorem completely general. The only contracted reduced symbol for odd t , with

$a = \frac{b}{2}$, is

$$2 \left| \begin{array}{ccc} 1 & & 1 \\ & 1 & \\ & 0 & \\ & \frac{t-1}{2} & \end{array} \right| t$$

yielding the true (if unsensational) fact that if t is odd, its quasi-order mod 2 is 1. Let us close with a more interesting numerical example.

Example: Let us take $t = 3$. The ψ -algorithm then proceeds by deciding, at each stage, whether $b - a$ or $b + a$ is divisible by 3 (remember that both a and b are prime to 3), and recording $\epsilon = 1$ in the first case, $\epsilon = 0$ in the second. It is unnecessary to record the value of q , since with $t = 3$, it is always 1. Thus, for example, we have symbols

$$37 \left| \begin{array}{cccccccc} 1 & 4 & 11 & 16 & 7 & 10 & 1 & \\ 2 & 1 & 1 & 1 & 1 & 3 & & \\ 1 & 1 & 0 & 1 & 1 & 1 & & \end{array} \right| \begin{array}{cccccccc} 2 & 13 & 8 & 5 & 14 & 17 & 2 & \\ 1 & 1 & 2 & 1 & 1 & 3 & & \\ 0 & 1 & 0 & 0 & 0 & 0 & & \end{array} \right|_3$$

each symbol shows that the quasi-order of 3 mod 37 is 9, and that $3^9 \equiv -1 \pmod{37}$. Despite some valid remarks in [2] on the relative complexity of the ψ -algorithm, we unrepentantly claim that it is more attractive to intelligent human beings than the φ -algorithm – though of course the φ -algorithm is certainly the more computer-friendly. We are not surprised at this dichotomy.

5 Folding All Star Polygons

In this section, we give an algorithm for introducing *secondary fold lines* (if necessary) into our tape, in order to be able by means of the FAT-algorithm, to construct any given regular $\{\frac{b}{a}\}$ -gon.

We assume that we can produce tape suitable for constructing (by the FAT-algorithm) any regular $\{\frac{b}{a}\}$ -gon *with a, b both odd, and $a < \frac{b}{2}$* . We first consider the case when we wish to construct a regular star $\{\frac{b}{a}\}$ -gon, where b is odd and a is even. Then $a = 2^k a'$ with a' odd, and we assert

Proposition 5.1: The tape prepared for folding the regular star $\{\frac{b}{a'}\}$ -gon has a crease line making an angle of $(\frac{a}{b})\pi$ with the forward direction of the top of the tape, and a second crease line under it making an angle of $(\frac{a}{b})\pi$ with it.

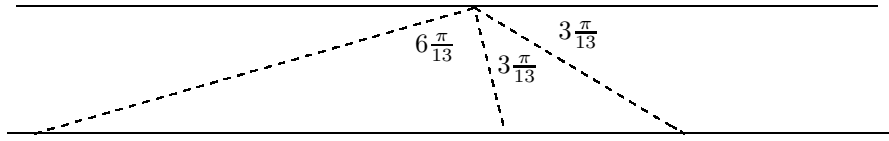


Illustration of Proposition 5.1

Figure 5.1

$$a = 6, \quad a' = 3, \quad k = 1, \quad b = 13$$

Proof: Our folding algorithm always starts with an *obtuse* angle, and proceeds by repeated bisection. Thus the angle of $\pi \frac{a'}{b}$ on the tape was created by successive bisection, by folding, of an angle $\frac{2^\ell \pi a'}{b}$ where

$$\frac{2^\ell \pi a'}{b} > \frac{\pi}{2}$$

Since $(\frac{2^k \pi a'}{b}) < \frac{\pi}{2}$, it follows that $k < \ell$, so that the bisection process must have produced the stated configuration.

Thus, in this case, we can simply use the FAT-algorithm, on the appropriate crease lines of the $\{\frac{b}{a'}\}$ -tape, to construct the desired $\{\frac{b}{a}\}$ -gon; no secondary folds are needed.

We now consider the problem of constructing a regular star $\{\frac{b}{a}\}$ -gon, where b is even and a is odd. Then $b = 2^k b'$, with b' odd. Let a be written in base 2 as

$$a = \epsilon_0 \epsilon_1 \cdots \epsilon_n, \quad \epsilon_0 = 1, \quad \epsilon_i = 0 \text{ or } 1, \quad 1 \leq i \leq n \text{ (in fact, } \epsilon_n = 1 \text{)} \quad (5.1)$$

We distinguish two possibilities. First, let $b' = 1$, so that $b = 2^k$. Since $2^n \leq a \leq 2^{k-1}$, we have $n \leq k - 2$. We initiate our algorithm with tape folded

to produce a crease line making an angle of $\frac{\pi}{2^{k-n}}$ with the forward direction of the top of the tape, with a second crease line under it making an equal angle with it. Second, let $b' \neq 1$. If $k > n$, start with the b' -tape and introduce, by folding $(k - n)$ secondary crease lines, a crease line making an angle of $\frac{\pi}{2^{k-n}b'}$ with the forward direction of the top of the tape, with a second crease line under it making an equal angle with it. If $k \leq n$, then the b' -tape will already have on it a crease line making an angle of $\frac{\pi}{2^{k-n}b'} (= \frac{2^{n-k}\pi}{b'})$ with the forward direction of the top of the tape, with a second crease line under it making an equal angle with it (see the proof of Proposition 5.1).

We have now described the *initial configuration*, consisting of two crease lines, the first making an angle of α_0 with the top of the tape, the second making an angle σ_0 with the first. We call α_0 the 0th *proximand* and σ_0 the 0th *support*, and proceed to define inductively the i th *proximand* α_i and the i th *support* σ_i , $0 \leq i \leq n$. Of course, $\alpha_0 = \sigma_0 = \frac{\pi}{2^{k-n}b'}$.

Suppose α_i, σ_i already defined, $i < n$. Bisect σ_i by a new crease line, into two equal angles λ_i, ρ_i , with λ_i to the left of ρ_i . If $\epsilon_{i+1} = 0$, define $\alpha_{i+1} = \alpha_i$, $\sigma_{i+1} = \rho_i$. If $\epsilon_{i+1} = 1$, define $\alpha_{i+1} = \alpha_i \cup \rho_i$, $\sigma_{i+1} = \lambda_i$. See Figure 5.2.

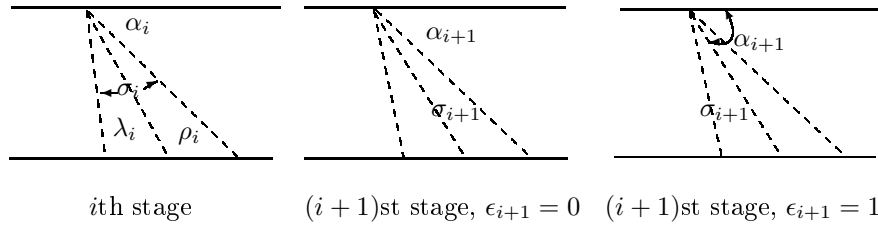


Figure 5.2

Proposition 5.2: The n th proximand α_n yields the required angle of $(\frac{a}{b})\pi$.

Proof: We argue by induction on i that

$$\alpha_i = \frac{\pi a_i}{2^k b'}, \quad \sigma_i = \frac{\pi}{2^{k-n+i} b'} \quad (5.2)$$

where

$$a_i = \epsilon_0 \epsilon_1 \cdots \epsilon_i 0 \cdots 0 \text{ (length } (n+1)) \quad (5.3)$$

Then (5.2) clearly holds for $i = 0$, since $a_0 = 2^n$. Given (5.2) and $\epsilon_{i+1} = 0$, then $\alpha_{i+1} = \alpha_i = \frac{\pi a_{i+1}}{2^k b'}$, since $a_{i+1} = a_i$, and $\sigma_{i+1} = (\frac{1}{2})\sigma_i = \frac{\pi}{2^{k-n+i+1} b'}$. Given (5.2) and $\epsilon_{i+1} = 1$, then

$$\begin{aligned} \alpha_{i+1} &= \alpha_i + \frac{1}{2}\sigma_i = \frac{\pi a_i}{2^k b'} + \frac{\pi}{2^{k-n+i+1} b'} = \frac{\pi}{2^k b'}(a_i + 2^{n-i-1}) \\ &= \frac{\pi}{2^k b'}(\epsilon_0 \epsilon_1 \cdots \epsilon_i 0 \cdots 0 + 00 \cdots 010 \cdots 0) \end{aligned}$$

where the digit 1 is in place $(n - i)$ counting from the right, and thus in place $(i + 1)$ counting from the left. Thus

$$\alpha_{i+1} = \frac{\pi}{2^k b'} (\epsilon_0 \epsilon_1 \cdots \epsilon_i 1 0 \cdots 0) = \frac{\pi a_{i+1}}{2^k b'}; \quad \sigma_{i+1} = \frac{1}{2} \sigma_i = \frac{\pi}{2^{k-n+i+1} b'}$$

This establishes (5.2), and we see immediately that $a_n = a$, so that $\alpha_n = \frac{\pi a}{2^k b'} = \frac{\pi a}{b}$. Given the crease line on the tape making an angle of $\frac{\pi a}{b}$, we may plainly complete the FAT-algorithm to construct the regular star $\{\frac{b}{a}\}$ -gon. Figure 5.3 illustrates this process for the $\{\frac{10}{3}\}$ -gon. Figure 5.4 illustrates a $\{\frac{17}{6}\}$ -gon constructed from $\{1, 1, 2\}$ tape in accordance with Proposition 5.1.

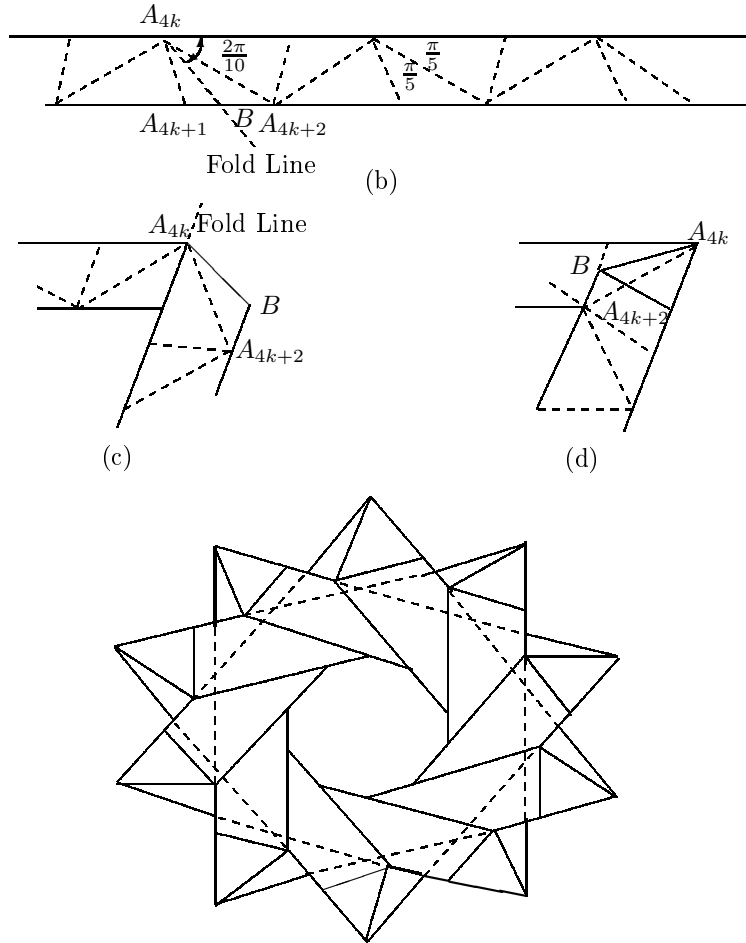


Figure 5.3

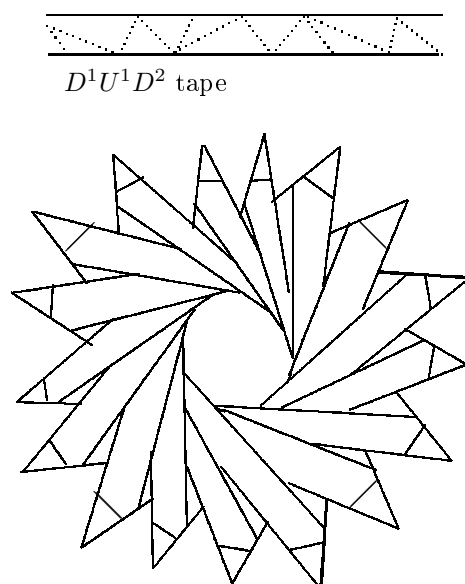


Figure 5.4

Remark: We have been concerned to give an algorithm; that is, a systematic procedure for folding regular $\{\frac{b}{a}\}$ -gons. Of course, we do not advocate this algorithm as the best procedure in any particular case. For example, if $a < (\frac{1}{2})b'$, it is plainly best to take the tape prepared for folding the $\{\frac{b'}{a}\}$ -gon, on which there will already be a crease line making an angle of $(\frac{a}{b'})\pi$ with the forward direction of the top of the tape, and then to bisect this angle by folding k times to produce the required angle of $(\frac{a}{b})\pi$. This remark may seem always applicable if $a = 1$, that is when regular *convex* polygons are in question; however, with $a = 1$, this simplified procedure actually coincides with our algorithm!

Appendix - The Further Generation Of Folding Numbers

In Section 3, we described the set of t -folding numbers, F_t , consisting of integers s of the form

$$s = \frac{t^{xy} - 1}{t^x - 1}, \quad x \geq 1, \quad y \geq 2 \quad (\text{A1})$$

We report here the main results of [10], in which we generalize even further, to consider the set F_{tu} of (t, u) -folding numbers s , being integers of the form

$$s = \frac{t^{xy} - u^{xy}}{t^x - u^x}, \quad x \geq 1, \quad y \geq 2 \quad (\text{A2})$$

In (A2), t, u are fixed positive integers with $t > u \geq 1$ and $\gcd(t, u) = 1$. We are content in this appendix simply to report the results; the proofs will be found in [10]. In each case, we indicate the result of Section 3 which is being generalized.

Theorem A1: If $\gcd(a, b) = d$, then $\gcd(t^a - u^a, t^b - u^b) = (t^d - u^d)$. (*Theorem 3.1*)

Now let c be a positive integer ≥ 2 , prime to t and u ; and let $\frac{F}{c}$ be the set defined by the rule

$$\frac{F}{c} = \{s \in F_{tu} \text{ such that } c \mid s\}$$

Then $\frac{F}{c}$ is non-empty; indeed, we may fix x in (A2) and the set of s of the form (A2) with a given x such that $c \mid s$ is non-empty.

We fix $x = 1$ in (A2), let y_0 be minimal such that $c \mid \frac{(t^{y_0} - u^{y_0})}{(t - u)}$, and write $y_0 = h(c)$, the *height* of c . Then

Lemma A2: If $c \mid \frac{(t^{xy} - u^{xy})}{(t^x - u^x)}$, then $h(c) \mid xy$. (*Theorem 3.4 (ii)*)

Theorem A3: Let s_0 be the smallest integer in $\frac{F}{c}$, where $s_0 = \frac{(t^{x_0 y_0} - u^{x_0 y_0})}{(t^{x_0} - u^{x_0})}$. Then $h(c) = x_0 y_0$. (*Theorem 3.5*)

Corollary A4: If there exist (t, u) such that $\frac{(t^{xy} - u^{xy})}{(t^x - u^x)} \mid \frac{(t^{x'y'} - u^{x'y'})}{(t^{x'} - u^{x'})}$, then $xy \mid x'y'$. (*Corollary 3.6*)

The main theorem of [10] is the following. Suppose s, s' are the two (t, u) -folding numbers

$$s = \frac{t^{xy} - u^{xy}}{t^x - u^x}, \quad s' = \frac{t^{x'y'} - u^{x'y'}}{t^{x'} - u^{x'}} \quad (\text{A4})$$

and suppose that $xy \mid x'y'$. Let $d = \gcd(xy, x')$, $h = \gcd(x, x')$, and set $xy = ad$, $x' = bd$. Then $ad \mid bdy'$, so that $a \mid by'$, hence $a \mid y'$.

Theorem A5: $s \mid s' \iff \frac{(t^d - u^d)}{(t^h - u^h)} \mid \frac{y'}{a}$. (Theorem 3.7)

This theorem provides an effective algorithm for deciding for what values of (t, u) we have $s \mid s'$. For there are 2 cases to be considered:

- (1) $\gcd(xy, x') \mid x$, i.e., $d = h$. Then $s \mid s'$ for all pairs (t, u) .
- (2) $\gcd(xy, x') \nmid x$, i.e., $h \mid d$, $h \neq d$. Then $\frac{(t^d - u^d)}{(t^h - u^h)} < \frac{y'}{a}$ for only finitely many pairs (t, u) , so that we may test for the required divisibility in a finite number of steps.

An easy computer algorithm answers our question quickly. With regard to Case 1, we have the following important refinement.

Theorem A6: For s, s' as in (A4), the following statements are equivalent:

- (i) $s \mid s'$ for all pairs (t, u) ;
- (ii) $s \mid s'$ for infinitely many pairs (t, u) ;
- (iii) $s \mid s'$ as polynomials in $Z[t]$ (or $Z[t, u]$);
- (iv) $xy \mid x'y'$ and $\gcd(xy, x') \mid x$

(Theorem 3.13)

Example: We may now extend an example given in Section 3. First, we find that the relation $7 \mid 21$ is a special case of the general fact that

$$\frac{t^3 - u^3}{t - u} \mid \frac{t^6 - u^6}{t^2 - u^2} \text{ for all pairs } (t, u)$$

Second, however, we find that the relation $3 \mid 21$ generalizes to the statement $\frac{(t^3 - u^3)}{(t - u)} \mid \frac{(t^6 - u^6)}{(t^2 - u^2)}$, which is *false* for *all* pairs (t, u) , with the single exception of $t = 2, u = 1$. No two generalizations could possibly have more divergent truth values!

More sophisticated, but very striking, is the following divisibility result involving integers in F_{tu} . Given any positive integer n , let $\psi(n)$ be the exponent of the multiplicative group $(\frac{\mathbb{Z}}{n})^*$ of residues prime to n (or units of the ring $\frac{\mathbb{Z}}{n}$). Consider the set S_q of all integers s of the form $s = \frac{(t^{xy} - u^{xy})}{(t^x - u^x)}$, such that $xy = q$.

Theorem A7: If $s \in S_q$, then $q \mid \psi(s)$. (Theorem 3.15 (ii) and the final note of Section 3)

We describe in [10] how to calculate $\psi(s)$. For example, let $q = 6$, $t = 3$, $u = 2$. Then

$$s = \frac{3^6 - 2^6}{3^2 - 2^2} = 133 \in S_6$$

Now $133 = 7 \times 19$, and $\Phi(7) = 6$, $\Phi(19) = 18$. Thus $\psi(133) = \text{lcm}(6, 18) = 18$, and $6 \mid 18$. We may also take $s = \frac{(3^6 - 2^6)}{(3^3 - 2^3)} = 35 \in S_6$, and $\Phi(5) = 4$, $\Phi(7) = 6$. Thus $\psi(35) = \text{lcm}(4, 6) = 12$, and $6 \mid 12$. These calculations show how fine Theorem A7 is, for $\gcd(18, 12) = 6$.

Postscript - Some Questions For Diligent Readers

2.1: What configurations can be obtained by taking tape folded by our systematic procedure and then applying some rule other than the FAT-algorithm? As special cases, consider the systematic procedures $D^n U^n$, $D^m U^n (m \neq n)$, $D^1 U^1 D^3$.

2.2: Suppose that regular convex $(2^n + 1)$ -gons are obtained from $D^n U^n$ -folded tape in various ways. How are these polygons related geometrically? (We know that there are at least $(n + 1)$ different types of regular $(2n + 1)$ -gon obtainable from this tape.)

2.3: Generalize Lemma 2.1, especially in ways relevant to our folding procedures. Here is one suggestion:

Let $\{x_{ik}\}$, $i = 1, 2, \dots, r$, $k = 0, 1, 2, \dots$, be r sequences of real numbers such that

$$\begin{aligned} x_{1k} + a_1 x_{2k} &= b_1, \\ x_{2k} + a_2 x_{3k} &= b_2, \\ &\vdots \\ x_{r-1,k} + a_{r-1} x_{rk} &= b_{r-1}, \\ x_{rk} + a_r x_{1,k+1} &= b_r \end{aligned}$$

for all k . Find conditions under which all the sequences converge. If the sequences converge, find their limits.

3.1: Find a proof of Theorem 3.1 based on the fact that $\gcd(u, v)$ is a linear combination of u and v with integer coefficients.

3.2: Let a be prime to t , and let $h(a)$ be the t -height of a . We saw that if a is prime, the smallest s in F_t with $a \mid s$ is given by

$$s = \frac{t^{h(a)} - 1}{t^{x_1} - 1}$$

where x_1 is the largest proper factor of $h(a)$. Generalize this result by weakening the hypothesis that a is prime.

3.3: How many regular star polygons with N vertices are there? If N is a folding number, how many of these may be obtained from tape folded to produce the regular convex N -gon?

4.1: Prove that (4.3) yields the unique solution to equation (4.2).

4.2: Let S be any set. Show that the sequence $(k_1, k_2, \dots, k_{r-1})$, where $k_i \in S$, may be extended in at most one way to a repeating sequence $(k_1, k_2, \dots, k_{r-1}, k_r)$, with $k_r \in S$. Here the sequence $(k_1, k_2, \dots, k_r)$ is said to *repeat* if there exists a proper divisor s of r such that $k_j = k_{s+j}$ for all j . What is the relevance of this to the content of Section 4?

- 4.3:** Give an algorithm for determining what star polygons are available from tape folded according to the sequence of instructions $(k_1, k_2, \dots, k_r)$. Here of course, the k_i are positive integers, and you may assume that the sequence is non-repeating.
- 5.1:** Describe explicitly how to fold a regular $\{\frac{60}{7}\}$ -gon by the algorithm of Section 5.
- 5.2:** Give a formula for the number of secondary fold-lines to be put into our (already folded) tape to produce a regular star $\{\frac{b}{a}\}$ -gon, where b is even.
- A.1:** Prove Theorem A1. (Hint: Look at Question 3.1.) What can be said if $\gcd(t, u) \neq 1$?
- A.2:** State and prove the generalization of Lemma 3.10 needed in the proof of Theorem A5.
- A.3:** Let $\gcd(t, u) = 1$. Define the equalizer of $t, u \pmod b$ (where $\gcd(t, b) = \gcd(u, b) = 1$) to be the smallest positive integer n such that $t^n \equiv u^n \pmod b$. Define the *quasi-equalizer* of $t, u \pmod b$ similarly. Give an algorithm for determining the quasi-equalizer of $t, u \pmod b$, and for deciding whether $t^k \equiv u^k \pmod b$ or $t^k \equiv -u^k \pmod b$, where k is the equalizer.
(Hint: Base this on the algorithm of Section 4 for determining quasi-orders mod b .)

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